

MMORPG / Crime RPG Ultra Deep Simulation Smoke Test (Live Pass 2)

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Project: znvsyzstmxhqvdckmgdt

Scope: Backup-first execution + 1300 bot, 30-day live simulation + security/performance lint pass

1) Backup Gate Status

- In-DB full snapshot backup: PASS
- backup_id: `bkp\_20260613\_081424`
- backup_schema: `backup\_20260613\_081424`
- captured_tables: 74
- captured_rows: 43353
- External pg_dump all-schema: BLOCKED by Supabase role permissions (`auth` schema lock denied)
- External pg_dump public-schema: BLOCKED by table permission (`public.items` lock denied)
- Supabase CLI `db dump`: BLOCKED locally because Docker daemon is required and unavailable

Decision applied for continuation:

- Proceeded with in-DB full snapshot as rollback anchor.

2) Phase-1 Hardening Applied

Migration applied:

- `phase1\_security\_hardenening\_search\_path`
 - Action: set `search\_path` to `public, pg\_temp` for all `public` functions
- `phase1\_performance\_add\_missing\_fk\_indexes`
 - Action: auto-create missing FK-leading indexes in `public`

3) 1300 Bot / 30 Day Live Simulation Run

- Seed call: `select \* from public.qa\_seed\_bots(1300);`
 - Result: `seeded\_bots = 1300`
- Run call: `select \* from public.qa\_run\_30\_day\_simulation(30);`
 - Result:
 - success: true
 - run_id: `b82690d4-548e-43ef-a261-4cd03dfbdbc7`
 - days: 30
 - bots: 1300
 - events: 39000

4) Economy / Activity Metrics

From `qa\_sim\_daily\_events` and `qa\_sim\_checkpoints` for run `b82690d4-548e-43ef-a261-4cd03dfbdbc7`:

- total_events: 39000
- distinct_active_bots: 1300
- avg_session_minutes (active): 47.76
- total_gold_earned: 60321288
- total_gold_spent: 38825634
- total_gems_spent: 253798
- total_items_minted: 136145
- total_items_burned: 77727
- exploit_findings: 6

Economic pressure signal:

- net_gold_inflation = 60321288 - 38825634 = 21495654 (positive inflation)

5) Retention Curve (Checkpointed)

Checkpoint data:

- Day 1: active 1238 / retained 1300
- Day 3: active 1158 / retained 1158

- Day 7: active 1054 / retained 1054
- Day 14: active 834 / retained 834
- Day 30: active 538 / retained 538

Derived rates (base cohort 1300):

- D1 active rate: 95.23%
- D7 active rate: 81.08%
- D30 active rate: 41.38%

Interpretation:

- Mid-term decay is acceptable for synthetic load testing but Day-30 is low for target MMORPG retention goals.

6) Exploit Findings (Top Risk)

- critical: `multi\_account\_gold\_funnel`
 - attempts: 260
 - success: 83
 - impact: 1,850,000 gold + 1,200 gems
- critical: `premium\_stack\_abuse`
 - attempts: 90
 - success: 14
 - impact: 480,000 gold
- high: `market\_price\_manipulation`
 - attempts: 190
 - success: 71
 - impact: 1,240,000 gold
- high: `cooldown\_bypass`
 - attempts: 110
 - success: 19
 - impact: 320,000 gold + 170 gems
- high: `prison\_escape\_abuse`
 - attempts: 126
 - success: 46
 - impact: 260,000 gold + 710 gems
- medium: `hospital\_escape\_abuse`
 - attempts: 140
 - success: 52
 - impact: 620 gems

7) Advisor Snapshot After Phase-1

Security lints total: 416

- `anon\_security\_definer\_function\_executable`: 194
- `authenticated\_security\_definer\_function\_executable`: 194
- `rls\_disabled\_in\_public`: 21
- `security\_definer\_view`: 2
- `function\_search\_path\_mutable`: 2
- `rls\_enabled\_no\_policy`: 2
- `auth\_leaked\_password\_protection`: 1

Performance lints total: 360

- `unused\_index`: 162
- `no\_primary\_key`: 74
- `multiple\_permissive\_policies`: 60
- `auth\_rls\_initplan`: 58
- `duplicate\_index`: 6

Interpretation:

- Search-path hardening in `public` is complete, but remaining mutable search path issues are in `game` schema functions.
- Largest remaining risk is broad SECURITY DEFINER execution exposure and missing RLS coverage.

8) Regression Status

- DB migrations applied successfully: PASS
- 1300x30 simulation run completed: PASS
- Backup/rollback anchor present: PASS (in-DB snapshot)
- External offline backup artifact: FAIL (permission + docker constraints)

9) Immediate Remediation Queue

Priority-0:

- Restrict EXECUTE on SECURITY DEFINER functions from `anon` and `authenticated`; expose only vetted RPC wrappers.
- Enable RLS on all externally exposed `public` gameplay tables where disabled.

Priority-1:

- Add explicit policies on `game.user\_quests` and `public.mekan\_pvp\_matches` (currently RLS enabled but no policy).
- Remove SECURITY DEFINER from views unless strictly required.

Priority-2:

- Tune index set (drop unused/duplicate after query-plan validation).
- Add PKs to missing tables where data model allows stable identifiers.

10) Final Verdict

- Live smoke objective executed with larger cohort (1300 bots) and 30-day horizon.
- System is load-test runnable but NOT production-hardening complete due to critical security lint backlog and exploit success rates.
- Recommended release gate: BLOCK until Priority-0 is closed and revalidated with another 1300x30 run.